Applications of Artificial Intelligence
Unit 6 – Consolidation Activity

Activity 1: Prediction in Daily Life

Instance A: Time taken to complete a language course

1. Prediction Made: 6 months

2. Data/ experience applied:
   - Time taken to complete a similar course in the previous year.
   - Estimated completion time on the course prospectus

3. Confidence level: Medium.

4. Actual outcome: 7 months

5. Additional data:
   - Variances in difficulty level of the previous course
   - Other study and personal commitments arising during the 6-month period

Instance B: Time taken to travel to university by bus

1. Prediction: 75 minutes.

2. Data/ experience applied:
   - Travel time from the bus schedules
   - Previous travel times, based on time of day and day of the week.

3. Confidence level: High

4. Actual outcome: 90 minutes

5. Additional data:
   - Presence of roadworks on the route leading to delays
   - How frequently the bus will stop on the route

Instance C: Number of People to attend a social gathering

1. Prediction Made: 15 people will attend

2. Data/ experience applied:
    - Number of people invited
    - Day of the week
    - Time of day
    - Number of friends within the group with family commitments (e.g. young children)

3. Confidence level: Low

4. Actual outcome: 12 people.

5. Additional data:
    - Weather
    - Distance from each person’s home to the venue

Instance D: Time taken to line-dry laundry

1. Prediction Made: 4 hours

2. Data/ experience applied:
    - Types of clothes (material and weight)
    - The temperature indoors
    - Amount of time for which the radiator will be on
    - The weather outside

3. Confidence level: Medium

4. Actual outcome: 3 hours

5. Additional data:
    - Other room conditions: ventilation, airflow and humidity
    - Arrangement of clothing (e.g. number of items on each drying rack)

Instance E: Cost of furnishing a new apartment

1. Prediction Made: £750

2. Data/ experience applied:

- Size of apartment, number of rooms
- Furniture provided with the apartment
- Types and brands of furniture being purchased
- Whether each purchased item was new or pre-owned
- Number of people occupying the apartment

3. Confidence level: High

4. Actual outcome: £800

5. Additional data:
   - Availability of suitable pre-owned items for purchase
   - Delivery and assembly costs for larger items

Conclusion:
Similarly to Machine Learning, predictions in daily life often rely heavily on past data, which can be unreliable. Often factors not reflected in past data can affect outcomes.

Activity 2: Algorithm Selection

Scenario 1: Predicting house prices in a city

1. Type of task: **Regression** (Prices are a continuous value)

2. Suitable Machine Learning approach: **Supervised** (Using labelled training data such as present and historic price data.)

3. Recommended data features:
   - Location, age, size and build of the house
   - Neighbourhood crime rate,
   - Number of bedrooms,
   - Availability of desirable features such as garden and parking,
   - Presence of undesirable features such as noisy highways.

4. Potential challenges when implementing solutions:
   - **Concept drift:** Real-world house prices can change over time, e.g. due to political, social and economic factors)

- **Underfitting:** The model might be too simple to capture patterns, particularly as house prices can be affected by several different factors.

5. Evaluation methods for model performance
   - **Regression metrics**: these measure the spread of prediction errors around the actual value. E.g. MAE, RSME, MAPE.
   - **Testing recent house listings:** If the model was trained on past listings, testing outputs against recent listings is a way to measure effectiveness using current data.

<u>Scenario 2: Categorising customer complaints</u>

1. Type of task: **Classification** (Can be ranked on a fixed scale, e.g. a scale of 1 to 10)

2. Suitable Machine Learning approach: **Supervised**

3. Recommended data features:
   - Time of complaint,
   - Price of goods purchased,
   - Legal obligations to customers based on affected goods or services (e.g. warranties, consumer statutory rights).

4. Potential challenges when implementing solutions:
   - **Overfitting**: a model for complaints can quite easily be designed to be too complex, and learn old patterns, becoming unable to adapt to new patterns.
   - **Small dataset:** For smaller businesses, there might be too few complaints to effectively train a Machine Learning model.

5. Evaluation methods for model performance
   - **Human validation:** Experienced human workers can review the algorithm's classification and evaluate accuracy and appropriateness.
   - **Complaint resolution:** A well-performing model will increase the speed and quality of complaint resolution.

<u>Scenario 3: Detecting fraudulent credit card transactions</u>

1. Type of task: **Classification** (Classified as likely or unlikely to be fraudulent)

2. Suitable Machine Learning approach: **Supervised** (Using labelled training data, such as historic fraudulent and legal transactions)

3. Recommended data features: Value of transaction, merchant category

4. Potential challenges when implementing solutions:
   - **Underfitting**: Model may not successfully capture patterns, leading to actual fraudulent transactions not being flagged
   - **Concept drift:** Fraud patterns are constantly evolving, as technologies develop. and there will be difficulty in keeping the algorithm updated.

5. Evaluation methods for model performance:

   The following measures can be used:

   - Precision: The percentage of flagged transactions that were actually fraudulent
   - Recall: The percentage of fraudulent cases that the model catches
   - Accuracy: Percentage of total cases that the model classifies correctly

<u>Scenario 4: Recommending products to online shoppers</u>

1. Type of task: **Regression** (Weights can be used to numerically measure how likely the shopper is to be interested in each product)

2. Suitable Machine Learning approach: **Supervised** (Labelled training data such as customer reviews).

3. Recommended data features:
   - Age of customers,
   - Frequency of purchases.

4. Potential challenges when implementing solutions:
   - **User conflation:** If a single account is used by multiple users (e.g. family members sharing an account), this presents a challenge for distinguishing the users and providing relevant recommendations to each.

   - **Historical bias:** Biases in training data can negatively affect the effectiveness of the model. For example, demand might be markedly different based on seasons (e.g. Christmas) or one-off events (e.g. severe heatwaves and the COVID-19 pandemic)

5. Evaluation methods for model performance
   - **Collect and Review feedback** from customers through surveys.

   - **Assess effectiveness of recommendations:** For example, what percentage of recommendations lead to purchases?

Scenario 5: Identifying natural customer segments for marketing

1. Type of task: **Clustering** (Finds similarities where these exist, as assign customers to groups accordingly).

2. Suitable Machine Learning approach: **Unsupervised** (no clear labelled examples for the algorithm to learn from).

3. Recommended data features:
   - Age of customer
   - Frequency of purchases
   - Value of purchased goods.

4. Potential challenges when implementing solutions
   - **User conflation:** If a single account is used by multiple users (e.g. family members sharing an account), this presents a challenge for distinguishing the users and providing relevant recommendations to each.

   - **Underfitting**: As modern human populations are complex, and there are many possible ways of classifying customers, it is possible that the training data might be too narrow to be effective.

5. Evaluation methods for model performance
   - **Distinctiveness of segments**: An evaluation could consider whether the segments created by the algorithm are truly distinct.

   - **Effectiveness of campaigns**: Testing campaigns can be run using the segments created by the algorithm, and the effectiveness of these will reflect the effectiveness of the model.

<u>Activity 3: Data to Decision Mapping</u>

Explanatory notes

1. Sources of raw data in the domain
   - **Inventory data:** stock currently held, stock movement patterns.

   - **Demand data**: loyalty card data, shop foot traffic (with CCTV and infrared sensors).

   - **Future demand estimates:** Fashion trends, weather, competitor activity

2. How this data is processed and transformed
   - Small data sets: Traditional data processing software, such as spreadsheets
   - Larger data sets: Big Data-enabled software, Power BI or specialised retail software

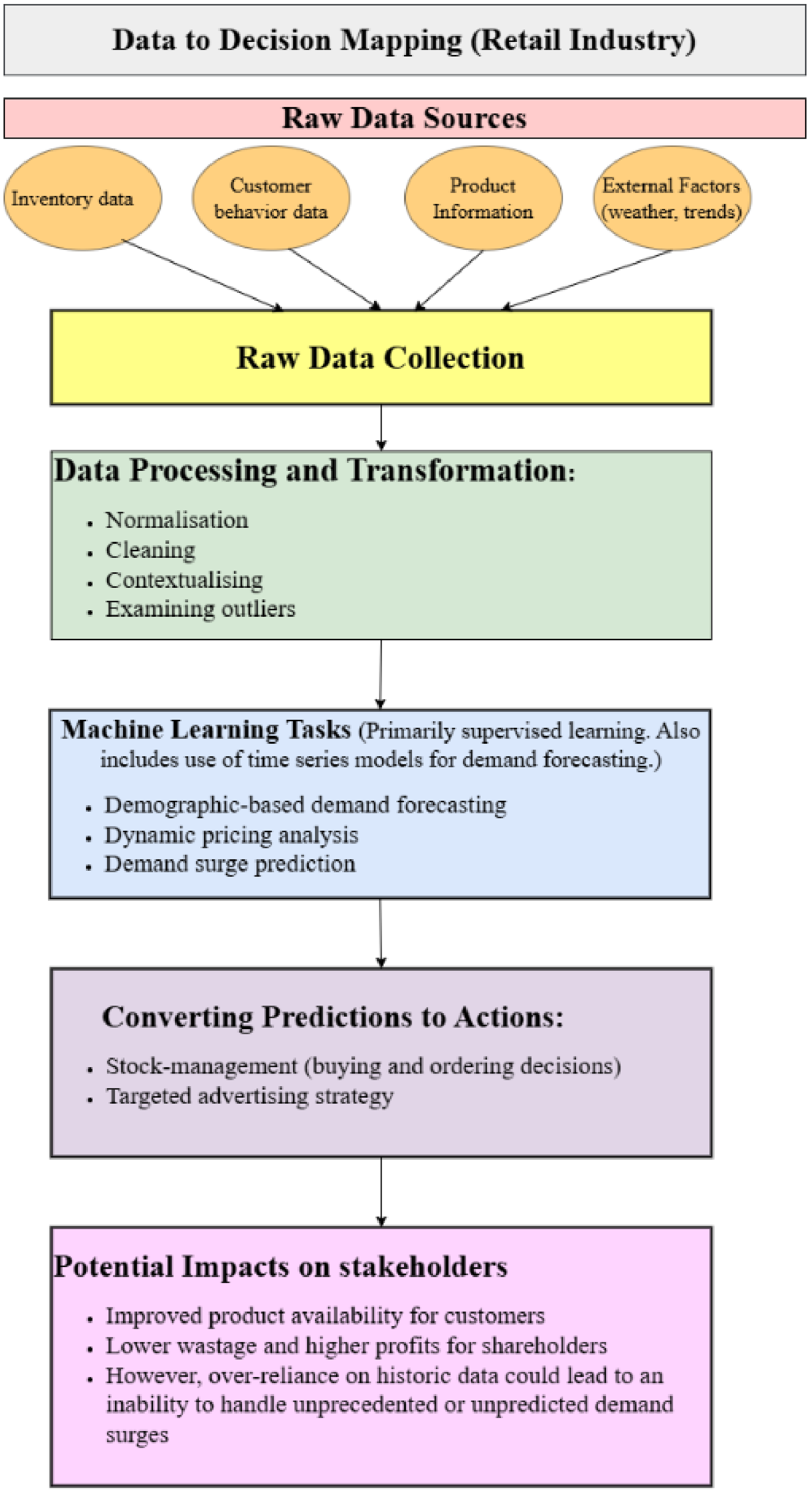
3. What machine learning task(s) would be applied
   - **Shelf monitoring:** Identify empty shelves in real-time and identify which products move off shelves faster at various times of the day.

   - **Footfall analysis and heat mapping:** Consider which shelves on the shopfloor are most frequently visited, and where customers spend the longest periods.

4. How predictions would be converted into actions
   - Stock management decisions
   - Targeted advertising

5. The potential impact of these actions on users and stakeholders
   - Lower wastage of inventory
   - Higher profits for shareholders

DECISION MAPPING FLOW CHART (RETAIL)



(Editing note- flowchart created on: https://app.diagrams.net/)

Sources

Google Developers. (2026). *Classification: Accuracy, recall, precision, and related metrics*. https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall